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Roll No

FT-703-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Heat Transfer, Combustion and Explosives

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What do you mean by study state heat conduction? Discuss different mode of heat transfer.
b) What is critical thickness of insulation on a small diameter wire or pipe? Explain its physical significance.
2. a) What are various hazards of smokes? Explain in brief.
b) Explain about smoke toxicity and evaluate any one model.
3. a) Explain Burning, Ignition and combustion phenomenon and discuss different types of flames.
b) Explain the chemical mechanism of combustion, also write its characteristics.
4. What do you mean by combustion. Explain the different modes of combustion.
5. Write short notes on following:
 - i) Plank's distribution law
 - ii) Intensity of Radiation

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6. How can be classify heat exchangers and fired heaters?
7.
 - a) Calculate the weight and volume of air needed for complete combustion of 5kg of coal containing 85% carbon, 10% hydrogen and rest oxygen.
 - b) State the Kirchhoff's law for radiation and describe the phenomenon of radiations from real surface.
8.
 - a) Explain the mechanism of detonation.
 - b) Write short notes on:
 - i) Classification of explosives
 - ii) Lower flammability point
 - iii) Upper flammability point.
